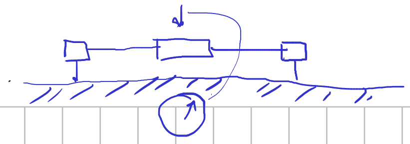
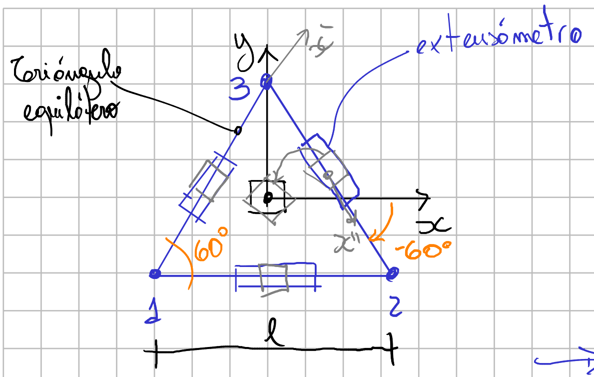
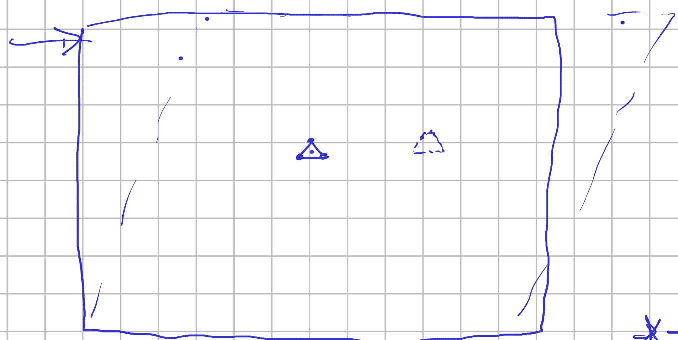
Pequeñas deformaciones





$\Delta l_{12}, \Delta l_{23}, \Delta l_{31} \rightarrow$ Datos

$$\varepsilon_{xx} = \frac{ds_x - ds_{0x}}{ds_{0x}} = \frac{\Delta l_{12}}{l_{12}} = \frac{\Delta l_{12}}{l}$$



$$\begin{cases} * \ \varepsilon_{xx} = \dfrac{\Delta l_{12}}{l} \\ \varepsilon_{x'x'} = \dfrac{\Delta l_{13}}{l} \\ \varepsilon_{x''x''} = \dfrac{\Delta l_{32}}{l} \end{cases}$$

$* \rightarrow 3$ ecuaciones

$\downarrow$

3 incógnitas $(\varepsilon_{xx}, \varepsilon_{yy}, \varepsilon_{xy})$

$$\underline{\underline{\varepsilon}} = \begin{bmatrix} \varepsilon_{xx} & \varepsilon_{xy} \\ \varepsilon_{xy} & \varepsilon_{yy} \end{bmatrix}$$

es conocido   incógnitas $\rightarrow 60°$

$$* \ \varepsilon_{x'x'} = \frac{\varepsilon_{xx} + \varepsilon_{yy}}{2} + \frac{\varepsilon_{xx} - \varepsilon_{yy}}{2} \cdot \cos(2\theta) + \varepsilon_{xy} \cdot \sin(2\theta) \qquad *_1$$

$$\varepsilon_{x''x''} = \frac{\varepsilon_{xx} + \varepsilon_{yy}}{2} + \frac{\varepsilon_{xx} - \varepsilon_{yy}}{2} \cdot \cos(2(-\theta)) + \varepsilon_{xy} \cdot \sin(2\cdot(-\theta))$$

$$* \ \varepsilon_{x''x''} = \frac{\varepsilon_{xx} + \varepsilon_{yy}}{2} + \frac{\varepsilon_{xx} - \varepsilon_{yy}}{2} \cdot \cos(2\theta) - \varepsilon_{xy} \cdot \sin(2\theta) \qquad *_2$$

$\cos(2\theta) = 1 - 2\sin^2\theta$

$1 - \cos(2\theta) = 2\sin^2\theta$

$\cos(2\theta) = 2\cos^2\theta - 1$

$1 + \cos(2\theta) = 2\cos^2\theta$

$$*_1 + *_2 \Rightarrow \varepsilon_{x'x'} + \varepsilon_{x''x''} = \varepsilon_{xx} + \varepsilon_{yy} + (\varepsilon_{xx} - \varepsilon_{yy}) \cdot \cos(2\theta)$$

$$= \varepsilon_{xx} \underbrace{(1 + \cos(2\theta))}_{2\cos^2\theta} + \varepsilon_{yy} \cdot \underbrace{(1 - \cos(2\theta))}_{2\sin^2\theta}$$

$\cos(60°) = 1/2$

$$\varepsilon_{yy} = \frac{\overset{\Delta l_{13}/l}{\varepsilon_{x'x'}} + \overset{\Delta l_{32}/l}{\varepsilon_{x''x''}} - 2\overset{\Delta l_{12}/l}{\varepsilon_{xx}} \cos^2\theta}{2\sin^2\theta} \quad \theta = 60° \Rightarrow \varepsilon_{yy} = \frac{2\Delta l_{13} + 2\Delta l_{23} - \Delta l_{12}}{3l}$$

$\sin(2 \cdot 60°) = \sqrt{3}/2$

$$*_1 - *_2 \Rightarrow \varepsilon_{x'x'} - \varepsilon_{x''x''} = 2\varepsilon_{xy} \cdot \sin(2\theta) \Rightarrow \varepsilon_{xy} = \frac{\varepsilon_{x'x'} - \varepsilon_{x''x''}}{2\sin(2\theta)}$$

$\Downarrow$

$$\varepsilon_{xy} = \frac{\Delta l_{13} - \Delta l_{23}}{\sqrt{3} \cdot l}$$